

Mihir Ravindra Adelkar

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Summary

Software Engineer with 3+ years building production AI systems and distributed **Java/Python** services. Designed **MCP-based LLM architecture** orchestrating tool use across heterogeneous data sources with full audit trails, replacing 3–5 day manual workflows with automated 5–10 second decisions. Led a 12-service monolith-to-microservices migration on AWS coordinating 20+ engineers, with 45%+ infra cost reduction and production-grade observability and security controls.

Skills

Languages	Java, Python, TypeScript, JavaScript, Go, SQL
AI & GenAI	Claude, RAG, MCP, Agentic Workflows, Prompt Engineering, Vector DBs, LangChain, Embeddings, NLP
Backend	Spring Boot, FastAPI, Django, Node.js (Express/NestJS), Microservices, REST, GraphQL, gRPC, SOA, Open API
Data	PostgreSQL, MySQL, MongoDB, Redis, Pinecone, Kafka, Elasticsearch, Schema Design, Query Optimization
DevOps	AWS, Azure, GCP, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD, Prometheus, Grafana

Professional Experience

Full-Stack Developer, KeelWorks Foundation — Boston, MA Aug 2025 – Present

- Built internal AI agents using **Claude API** for code scaffolding, test generation, and PR documentation; integrated into the dev workflow for a distributed engineering team.
- Developing workforce engagement platform with **React/TypeScript** frontend and **Node.js** microservices on PostgreSQL; established reusable component libraries and coding standards across the team.
- Deployed containerized services on **AWS EKS** with Docker/Terraform; built end-to-end **CI/CD** pipelines via GitHub Actions enabling same-day deploys with automated quality and security gates.

Software Engineer, Freelancing (Dabadu.ai) — Remote May 2023 – Sept 2023

- Resolved production defects across XRM automotive dealership CRM (**React/Node.js**) including real-time WebSocket and **Elasticsearch** search modules; restored reliable deployments for live dealership operations.
- Refactored legacy JavaScript to **TypeScript**; established **Cypress** E2E testing and **SonarCloud** quality gates in CI/CD, reducing production bugs by 20%.

Software Engineer, NeoSOFT Technologies — Mumbai, India Jun 2021 – May 2023

- Audited a 500K+ user real estate CRM running on a **Java/Spring Boot monolith** (400–500ms avg latency, 2s spikes); proposed and led ground-up rebuild to **12 microservices** with **gRPC** and **Kafka** on AWS, coordinating 20+ engineers over 6 months — 7x faster inter-service calls and 45%+ infra cost reduction via AWS rightsizing.
- Architected full-stack applications using **Java/Spring Boot**, **Python/Django**, **Node.js**, and **React/TypeScript** across CRM and analytics platforms serving 500K+ users; designed **REST/GraphQL APIs** with **Redis** caching reducing response times by 70%.
- Established **CI/CD** pipelines with Jenkins and GitHub Actions; achieved 85% test coverage through **unit and integration testing** with JUnit/Jest/Cypress; instrumented services with **Prometheus/Grafana** and New Relic, reducing MTTR from 25 to 15 min.
- Implemented **IAM/RBAC** with 50+ roles for multi-tenant isolation, encrypted data flows, and **security** across REST and gRPC.
- Built open-source boilerplates — **mf-shell** (Webpack Module Federation for micro-frontends) and **Console** (multi-tenant admin panel with RBAC) — adopted by 15+ internal teams as scaffolding standards.
- Mentored 8 developers** through code reviews, pair programming, and bi-weekly knowledge-sharing seminars in Agile sprints.

Projects

Prior Authorization AI System — Regulated-Domain LLM Pipeline *Python, FastAPI, Claude API, MCP, FHIR R4, Pinecone, Tesseract OCR, GCP*

- Designed **MCP-based agentic architecture** where the LLM orchestrates tool calls across data extractors, OCR pipelines, and policy retrievers; every tool invocation logged for a full **audit trail** suitable for regulated environments.
- Built **RAG pipeline** on **Pinecone vector DB** combining domain documents with coverage guidelines for LLM-based decisions; output includes **confidence scoring** and **citation traceability** so reviewers can validate every claim.
- Replaced 3–5 day manual reviews with automated decisions in **5–10 seconds** at 95%+ extraction accuracy on test dataset; deployed on **GCP** with HIPAA-style PHI handling, encryption, and access controls.

CVE Intelligence System — RAG-Powered Security Assistant *Go, Python, Kafka, Pinecone, Llama 3, AWS EKS, Terraform, Istio*

- Built multi-component AI pipeline coordinating **Kafka** ingestion, **Pinecone** retrieval, and **Llama 3** generation services — engineers query vulnerabilities in plain English; system returns answers with source citations.
- Designed **fault-tolerant** distributed microservices in Go with schema validation and queues for real-time CVE ingestion.
- Deployed on **AWS EKS** with Terraform IaC and **Istio service mesh** for mTLS; instrumented with **Prometheus/Grafana** **observability** and Jenkins CI/CD with automated alerting.

Education & Certifications

Master of Science in Software Engineering Systems — *Northeastern University, Boston, MA* Sept 2023 - May 2025

Relevant Coursework: Advanced Cloud Computing, Distributed Systems, Algorithms, Object-Oriented Design, Big Data Systems

Bachelor of Engineering in Computer Engineering — *University of Mumbai, Mumbai, India* Aug 2018 - Jun 2021

Certifications: Certified Java Enterprise Applications Developer (Oracle) | Scrum Fundamentals Certified | Foundational AI